Mohamed Alhaskir

Machine Learning Researcher | Research Engineering, Multimodal AI

Machine learning researcher and PhD candidate at RWTH Aachen University with experience designing, implementing, and evaluating end-to-end AI systems. Research spans multimodal LLM applications, deep learning for physiological time series, and predictive modeling. Experienced in reproducible ML experimentation, model evaluation, GPU-based research workflows, and developing AI systems for real-world applications. First-author publications in applied machine learning, with an emphasis on reliability, robustness, and rigorous experimental validation.

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RESEARCH & ENGINEERING EXPERIENCE

PhD Researcher, Institute of Medical Informatics, RWTH Aachen University 2022 – Present

Aachen, Germany

HaMoJo: Multimodal LLM System for Clinical Simulation Assessment | github.com/Mohamed-Alhaskir/HaMoJo

- Architected and implemented an end-to-end multimodal AI pipeline integrating speech recognition, speaker diarization, computer vision, and locally hosted LLMs to generate structured feedback from simulation recordings.
- Developed reproducible evaluation workflows incorporating rubric-grounded LLM scoring, programmatic validation, deterministic inference, and comparison against human assessments.

  Technologies: Python, Whisper, NeMo, MediaPipe, Qwen2.5

MoniPy: Wearable ECG Seizure Detection | github.com/rwth-imi/monipy

- Designed and implemented a deep learning pipeline combining physiological signal processing, heart-rate variability features, automated data-quality assessment, and seizure classification.
- Evaluated the method using leave-one-patient-out validation against expert-labeled seizure events in a three-center clinical study involving 3,278 hours of recorded ECG; the final quality-screened evaluation included 161 seizures from 38 patients.

  Methods: Deep learning, signal processing, time-series analysis, model evaluation

FORECAST: Predictive Modeling for Hospital Patient Flow (ongoing)

- Developing machine learning models using routine emergency department data to predict inpatient ward admission and subsequent length of stay.
- Investigating predictive modeling approaches for emergency department crowding and hospital resource utilization.

Master's Researcher, Forschungszentrum Jülich (IBI-3) 2021 – 2022

- Master's thesis on spike variation in 3D microelectrode array recordings; developed signal-processing algorithms and optimized the SpyKING CIRCUS spike-sorting pipeline for characterizing spike-waveform variability.

Research Internships, Charité Berlin and RWTH Aachen 2020 – 2021

- Charité Berlin, AG Klinische Neurotechnologie: signal-acquisition and decoding algorithms for an OPM-MEG-based brain-computer interface.
- RWTH Aachen, Molecular & Systemic Neurophysiology: classification algorithms for neural decoding of wide-field calcium-imaging data.

TECHNICAL SKILLS

Programming	Python, SQL, Bash, MATLAB
Machine Learning	PyTorch, TensorFlow/Keras, scikit-learn, Hugging Face Transformers, LoRA fine-tuning
LLMs & Multimodal AI	vLLM, llama.cpp, Whisper, speaker diarization, MediaPipe, OpenCV
ML Evaluation	Subject-wise validation, model performance evaluation, reproducible experimentation, data-quality assessment
ML Engineering	Git, pytest, Docker, MLflow, Weights & Biases, FastAPI
Computing	Slurm, HPC environments, GPU-based experimentation
Data & Statistics	pandas, NumPy, SciPy, statsmodels, SQLAlchemy/MySQL
Languages	Arabic (native), German (C2), English (C1)

SELECTED PUBLICATIONS

Alhaskir, M., et al. *Reliable Detection of Focal Onset Impaired Awareness Seizures in Patients with Epilepsy Using Wearable ECG: Development and Validation Study.* Computer Methods and Programs in Biomedicine, 279, 109295, 2026.

Alhaskir, M., et al. *Feasibility of LLM-Assisted Communication Feedback in Simulation-Based Medical Education: A Pilot Study.* Studies in Health Technology and Informatics, 336, 2140–2144, 2026.

Tayebi Arasteh, S., Farajiamiri, M., Lotfinia, M., Hinrichs-Puladi, B., Bienzeisler, J., **Alhaskir, M.**, et al. *The Role of Self-Supervised Pretraining in Differentially Private Medical Image Analysis.* arXiv:2601.19618, 2026.

Alhaskir, M., et al. *ECG Matching: An Approach to Synchronize ECG Datasets for Data Quality Comparisons.* Studies in Health Technology and Informatics, 307, 225–232, 2023.

Full publication list: Google Scholar

EDUCATION

PhD, Medical AI, RWTH Aachen University — 2022 – 2027 (expected)
Institute of Medical Informatics · Aachen, Germany

MSc, Biological Information Processing, RWTH Aachen University — 2019 – 2022
Major: Biophysics · Aachen, Germany

BSc, Biological Information Processing, RWTH Aachen University — 2016 – 2019
Aachen, Germany

AWARDS & LEADERSHIP

DAAD Prize for Outstanding International Students, RWTH Aachen, 2021

Co-Founder, NeuroTX Aachen e.V., 2020–2022

Developed a brain-computer interface prototype for electric wheelchair control; project funded by the RWTH Aachen Collective Incubator.